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Garden of Knowledge and Virtue

MECHATRONICS SYSTEM INTEGRATION

MCTA 3203

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SECTION 1

WEEK 7: PLC INTERFACING WITH MICROCONTROLLER AND PC OVER ETHERNET

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ABSTRACT

This experiment explores the integration of a Programmable Logic Controller (PLC) with an Arduino microcontroller using the OpenPLC Editor to gain a better understanding of basic ladder logic programming and hardware interfacing. The experiment aimed to design, simulate, and upload ladder logic programs, as well as develop a functional Start–Stop control circuit using push-button switches and an LED. The procedure included creating a simple LED blinking program, running a simulation, configuring physical input and output addresses, and uploading the compiled program to the Arduino board. Afterwards, the Start–Stop circuit was assembled and tested to verify proper latching operation. The results demonstrated that OpenPLC is a dependable platform for developing PLC logic and executing it on hardware when the addressing and connections are configured correctly. In conclusion, open-source PLC systems are capable of demonstrating industrial automation concepts effectively and are highly beneficial for educational purposes.

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1.0 INTRODUCTION

A Programmable Logic Controller (PLC) is a specialized industrial computer designed to control and automate electromechanical processes in manufacturing and machinery. This experiment utilizes the OpenPLC Editor, an open-source software compliant with the IEC 61131-3 standard, to interface a personal computer with a microcontroller. The primary objective is to develop, simulate, and implement control logic, specifically ladder diagrams onto an Arduino-based system to manage basic automation tasks like LED blinking and start-stop control circuits.

2.0 MATERIALS AND EQUIPMENTS

- Software: OpenPLC Editor.
- Microcontroller: Arduino Uno
- Input Components: 2 Push Button Switches.
- Output Components: 1 LED .
- Circuit Components:
 - 10Ohm Resistors (for push buttons).
 - 560Ohm Resistor (for LED).
 - Breadboard and Jumper Wires.

3.0 EXPERIMENTAL SETUP

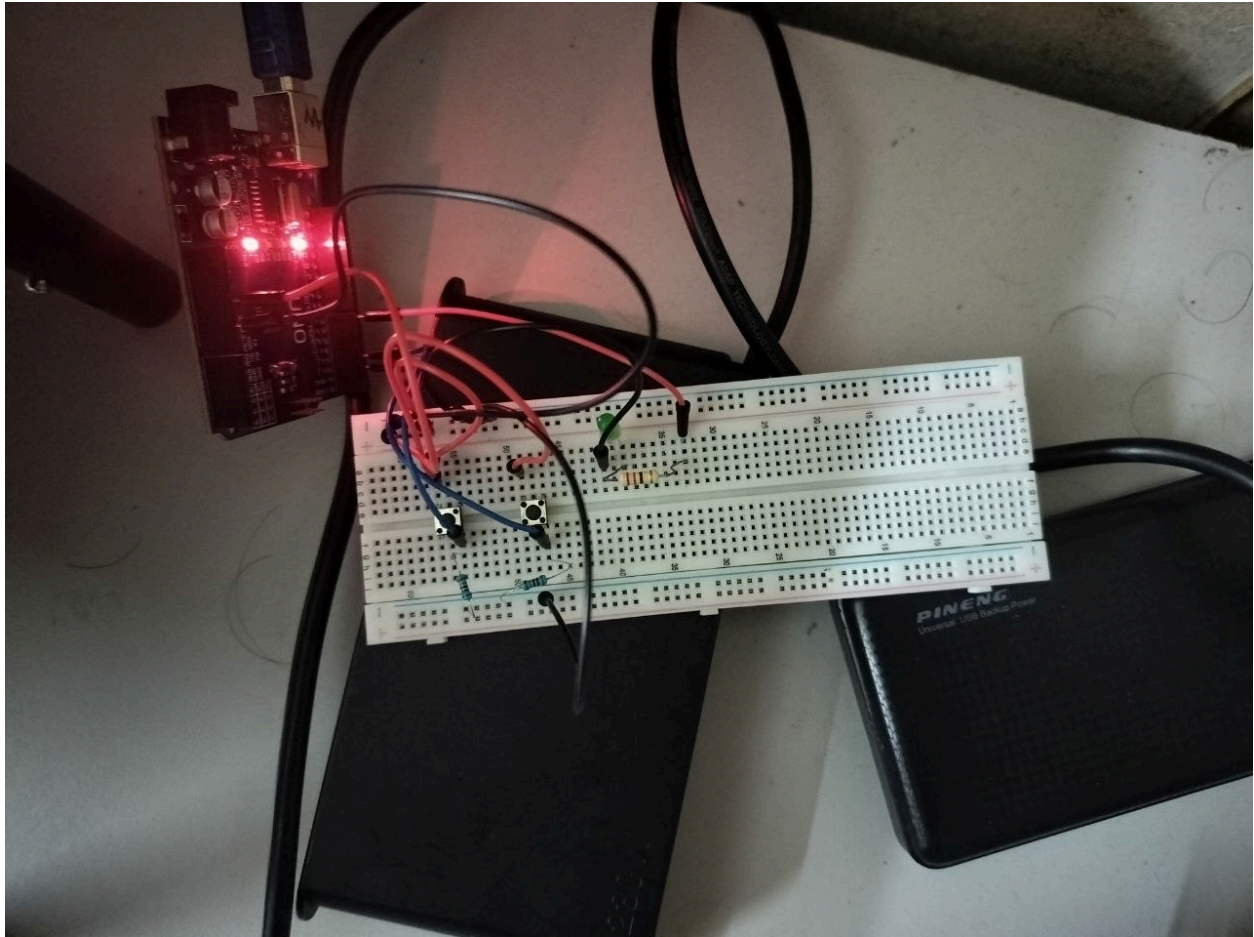
Software Configuration

1. Project Initialization: A new project is created in OpenPLC Editor, selecting Ladder Diagram (LD) as the programming language.
2. Variable Definition: Variables are defined with specific types (e.g., **BOOL**) and associated with physical addresses (Locations) corresponding to the Arduino pins.
 - Inputs: Assigned to **%IX** addresses (e.g., **%IX0.0** for Pin 2).
 - Outputs: Assigned to **%QX** addresses (e.g., **%QX0.0** for Pin 7).
3. Logic Construction: The ladder diagram is built using power rails, contacts (normally open or negated), and coils. For advanced timing, TON (Timer On-Delay) and TOF (Timer Off-Delay) blocks are integrated.

Hardware Integration

1. Circuit Assembly: The push buttons are wired to the Arduino digital input pins with pull-down resistors, and the LED is connected to a digital output pin through a current-limiting resistor.
2. Deployment: The Arduino is connected via USB. After selecting the correct board type and COM port, the compiled ladder logic is transferred to the PLC/microcontroller.

Setup Diagram



4.0 METHODOLOGY

4.1 Task 1: Blinking LED Ladder Diagram

- **OpenPLC Software Setup**

The OpenPLC Editor software was first downloaded and installed on the computer. After installation, the software was launched and a new project folder was created. A new Ladder Diagram (LD) project was then created within the software.

- **Ladder Diagram Development**

A Boolean variable was created in the OpenPLC Editor to represent the LED output. The ladder diagram was developed by inserting the left and right power rails, a negated contact, and an output coil according to the blinking LED circuit design provided in the experiment manual.

A Timer block (TON) was then added to the ladder diagram to control the blinking interval of the LED. A time delay expression was inserted to determine the blinking duration of the LED output.

- **Program Compilation and Simulation**

After completing the ladder diagram, the program was compiled in the OpenPLC Editor to ensure that no syntax or logic errors were present. The simulation function in OpenPLC Editor was then used to observe the LED blinking operation before transferring the program to the hardware.

- **Arduino Connection and Variable Addressing**

The Arduino board was connected to the computer using a USB cable. The OpenPLC variable was associated with the correct Arduino digital output pin using the physical addressing configuration provided in the OpenPLC documentation.

The correct Arduino board type and COM port number were selected before transferring the ladder diagram program to the Arduino board.

- **Program Upload and Testing**

The ladder diagram program was uploaded to the Arduino board using the OpenPLC Editor. After successful uploading, the LED connected to the Arduino output pin was observed blinking continuously according to the timer interval configured in the ladder diagram.

4.2 Task 2: Start-Stop Control Circuit

- **Ladder Diagram Design**

A Start-Stop control ladder diagram was created according to the circuit shown in Fig. 5 of the experiment manual. Variables representing the START push button, STOP push button, and LED output were defined in the OpenPLC Editor as Boolean variables.

The START push button was configured as a normally open contact, while the STOP push button was configured as a normally closed contact in the ladder diagram. An output coil was added to represent the LED output.

- **Variable Configuration and Address Mapping**

The input and output variables were mapped to the corresponding Arduino digital input and output pins using the OpenPLC physical addressing configuration. The correct pin associations between the OpenPLC variables and the Arduino board were verified before compilation.

- **Program Compilation and Simulation**

The ladder diagram was compiled in the OpenPLC Editor to verify proper logic operation. The simulation feature was then used to observe the Start-Stop control behavior before transferring the program to the Arduino hardware.

- **Hardware Circuit Assembly**

The circuit was assembled on a breadboard using an Arduino board, two push button switches, an LED, resistors, and jumper wires. The START push button was connected as a normally open input, while the STOP push button was connected as a normally closed input. The LED was connected to the designated Arduino digital output pin through a current-limiting resistor.

All circuit connections were checked carefully before powering the system to avoid incorrect wiring or short circuits.

- **Experimental Testing**

The compiled ladder diagram program was uploaded to the Arduino board through the OpenPLC Editor. After the upload process was completed, the functionality of the Start-Stop circuit was tested.

When the START push button was pressed, the LED turned ON and remained ON through the holding circuit logic. When the STOP push button was pressed, the LED turned OFF and the output operation stopped. The experiment was repeated several times to ensure stable and reliable operation of the control system.

5.0 DATA COLLECTION

During this experiment, data collection focused on observing and documenting the operational behavior of the Start–Stop control circuit implemented using OpenPLC, Arduino, and external push buttons. The following data were recorded:

5.1 Ladder Diagram Structure

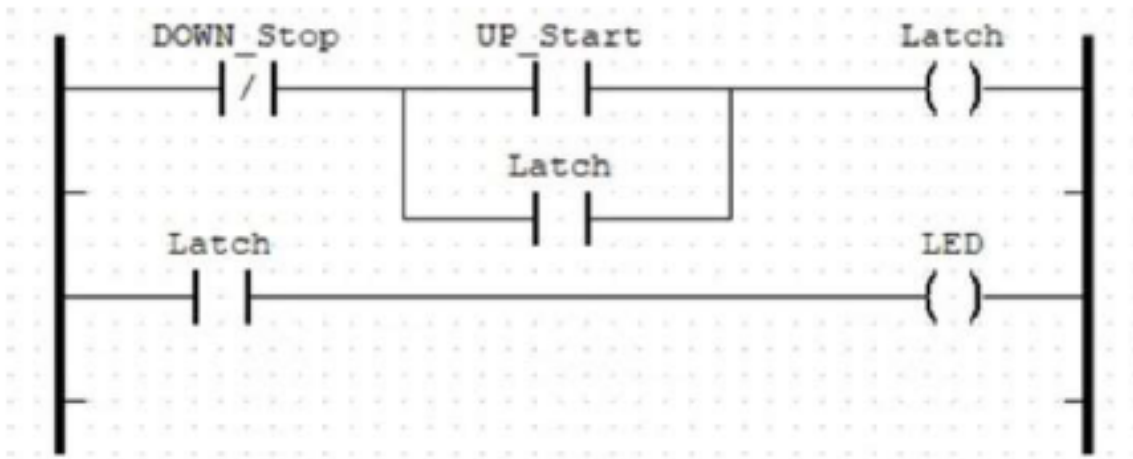


Figure 5.1 Ladder diagram used for Start-Stop control circuit

- START button input → %IX0.0
- STOP button input → %IX0.1
- OUTPUT coil (LED) → %IX0.0
- Internal latch variable → a BOOL variable used for latching the start signal

5.2 Data Recording Table

Trial	START Button	STOP Button	LED Output	Observation
1	Pressed	Not Pressed	ON	LED remained ON
2	Not Pressed	Pressed	OFF	LED turned OFF
3	Pressed	Released	ON	Holding circuit operated correctly
4	Repeated Operation	Repeated Operation	ON/OFF	Stable system response

5.3 Physical Testing Data

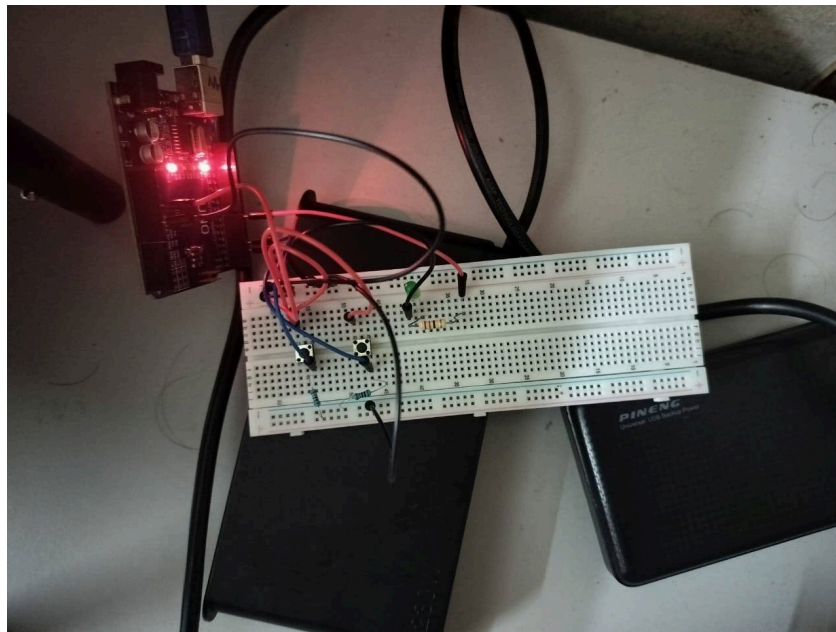


Figure 5.3 Hardware setup

Once uploaded to the Arduino:

- Pressing the START button turned the LED ON
- LED remained ON due to latch even after releasing the START button

- Pressing the STOP button turned the LED OFF
- LED remained OFF until START was pressed again

These observed responses (LED ON/OFF states) served as the primary collected data for evaluating the correctness of the control logic.

6.0 DATA ANALYSIS

Data collected from both the simulation environment and physical hardware were analyzed to determine whether the Start–Stop circuit functioned according to industrial PLC logic.

6.1 Task 1: Blinking LED Ladder Diagram Timer Function Analysis

- The Timer ON Delay (TON) block successfully controlled the LED blinking interval in the ladder diagram program.
- Different timer delay values produced different blinking speeds of the LED.
- Smaller timer values caused the LED to blink faster, while larger timer values produced slower blinking intervals.
- The experiment demonstrated that the timer block can effectively control output timing operations in PLC ladder logic systems.

LED Blinking Performance Analysis

- The LED blinking operation remained stable throughout the experiment without interruption.
- The OpenPLC simulation produced the same behavior as the actual Arduino hardware implementation, confirming correct ladder logic operation.
- The synchronization between the OpenPLC Editor and Arduino board was successful, allowing accurate transfer of the ladder program.

6.2 Task 2: Start-Stop Control Circuit

Start-Stop Logic Analysis

- The Start-Stop control circuit operated according to the designed ladder logic.
- When the START push button was pressed, the LED turned ON and remained ON due to the holding circuit (latching) configuration.
- The holding circuit allowed the output coil to stay energized even after the START push button was released.
- When the STOP push button was pressed, the circuit connection was interrupted and the LED turned OFF immediately.

System Response Analysis

- The response of the LED output to the push button inputs was immediate and stable throughout the experiment.
- No significant delay or malfunction was observed during repeated testing of the Start-Stop control circuit.
- The ladder diagram simulation in OpenPLC matched the actual hardware behavior on the Arduino board, proving the reliability of the system design.

7.0 RESULTS

The experiment yields three distinct functional outcomes based on the implemented logic:

Case 1: Basic Blinking LED

The initial ladder diagram consists of a single negated contact and a coil. When uploaded, the LED oscillates between ON and OFF states based on the PLC scan cycle.

Case 2: Controlled Time Interval

By introducing TON and TOF timer blocks (as seen in Fig. 3), the blinking frequency is precisely controlled.

- Observation: Setting the pulse time to **T#500ms** results in a steady, visible blink at half-second intervals.

Case 3: Start-Stop Control Circuit

This configuration implements a latching mechanism.

- Start Action: Pressing the "UP_Start" button (connected to **%IX0.0**) energizes the "Latch" variable and the LED.
- Latching: Once energized, the "Latch" contact in parallel with the Start button maintains the circuit even after the button is released.
- Stop Action: Pressing the "DOWN_Stop" button (negated contact at **%IX0.1**) breaks the circuit, de-energizing the LED.

8.0 DISCUSSION

This experiment demonstrated that ladder logic traditionally used in industrial PLCs can be implemented on a microcontroller platform using OpenPLC. By integrating software simulation with actual hardware, students gained a practical understanding of PLC programming, timing logic, and control circuits.

The success of the Start-Stop circuit indicates that even simple Arduino boards can replicate the behavior of industrial PLCs for educational purposes.

Some potential issues include:

- Incorrect COM port selection
- Wrong physical addressing for Arduino pins
- Loose wiring on breadboard
- Timer preset values entered incorrectly
- Unstable power supply

These issues were resolved through systematic checking and reconfiguration.

9.0 CONCLUSIONS

In conclusion, the experiment successfully demonstrated the interfacing of a Programmable Logic Controller (PLC) system with an Arduino microcontroller using the OpenPLC Editor platform. The objectives of developing, simulating, and implementing ladder logic control circuits were achieved successfully through both the blinking LED and Start–Stop control experiments. The Timer ON Delay (TON) and Timer OFF Delay (TOF) functions effectively controlled the LED blinking intervals, proving that timing operations can be implemented accurately using PLC ladder logic. In addition, the Start–Stop control circuit demonstrated proper latching operation, where the LED remained energized after the START push button was released and de-energized immediately when the STOP push button was pressed. The simulation results in OpenPLC matched the actual hardware behavior observed on the Arduino system, confirming the reliability and accuracy of the ladder logic implementation. The experiment also provided practical understanding of PLC programming concepts, physical input/output addressing, timer functions, hardware interfacing, and industrial-style control systems using a low-cost microcontroller platform. Overall, the experiment enhanced knowledge of automation control systems and demonstrated that Arduino-based platforms can effectively emulate basic industrial PLC operations for educational and prototyping purposes.

10.0 RECOMMENDATIONS

Several improvements can be considered to further enhance the effectiveness and functionality of future PLC interfacing experiments. Firstly, additional industrial sensors such as proximity sensors, limit switches, or temperature sensors can be integrated into the system to simulate more realistic automation processes and expand the complexity of the ladder logic program. More advanced PLC functions such as counters, sequential control, memory registers, and analog input/output operations may also be implemented to provide deeper understanding of industrial automation systems. To improve system reliability, proper PCB connections or terminal blocks could replace breadboard wiring, as loose jumper wire connections may affect signal stability during operation. Furthermore, using external power regulation and protection circuits would improve the safety and durability of the hardware components. Future experiments may also include Human Machine Interface (HMI) integration or SCADA monitoring systems to allow real-time visualization and control of the PLC process. In addition, improving troubleshooting procedures and documentation of physical addressing configurations would help reduce common programming and hardware connection errors during implementation. Overall, these improvements would enhance both the practical learning experience and the industrial relevance of the experiment.

11.0 REFERENCES

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APPENDICES

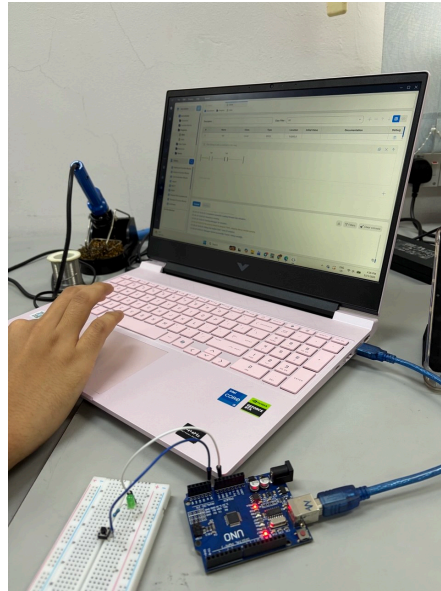


Figure 1.0: Experimental setup of circuit and OpenPLC Application

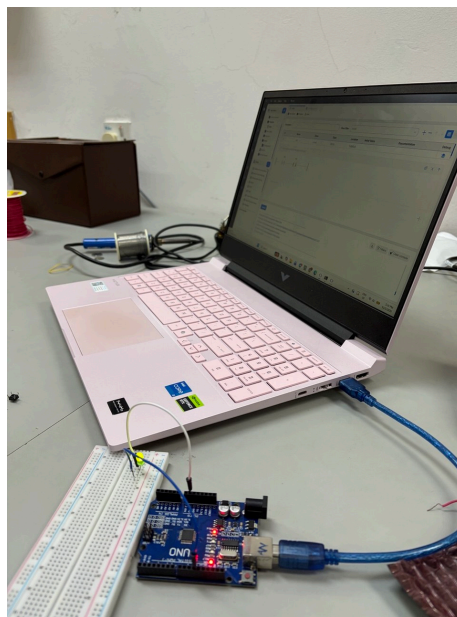


Figure 1.1: Uploading Coding to the Arduino Uno

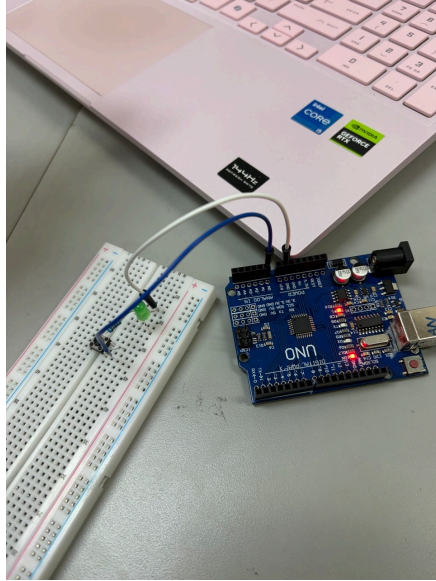


Figure 1.2: Testing the circuit of experiment

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STUDENTS' DECLARATION

Certificate of Originality and Authenticity

This is to certify that we are **responsible** for the work submitted in this report, that **the original work** is our own except as specified in the references and acknowledgment, and that the original work contained herein has not been undertaken or done by unspecified sources or persons. We hereby certify that this report **has not been done by only one individual**, and **all of us have contributed to the report**. The length of contribution to the reports by each individual is noted within this certificate. We also hereby certify that we have **read** and **understand** the content of the total report, and no further improvement on the reports is needed from any of the individual contributors to the report. We therefore agreed unanimously that this report shall be submitted for **marking**, and this **final printed report** has been **verified by us**.

Signature:  Name: Amir Aiman Bin Ariff Affendi Matric Number: 2315925 Contributions: Introduction, Materials and Equipment, Experimental setup, Results	Read [/] Understand [/] Agree [/]
Signature:  Name: Anis Aqilah Binti Abdullah Matric Number: 2329678 Contributions: Abstract, Conclusion, Recommendations, References, Appendices, Acknowledgements	Read [/] Understand [/] Agree [/]
Signature:  Name: Azril Ulwan Bin Mohammad Imri Matric Number: 2314617 Contributions: Methodology, Data Collection, Data Analysis, Discussions	Read [/] Understand [/] Agree [/]